Optimizing Inventory with Data

Forecasting Analysis

Agenda

Objective: To identify the most accurate forecasting model for NovaTech's sales data.

Methodology: Comparison of five forecasting models using one year sales data.

Focus: Analyzing data at two different intervals (Daily vs Monthly) to see which model fits best.

About NovaTech Store

NovaTech Store is a reputable retailer specializing in Lenovo Pcs (high-performance workstations and laptops).

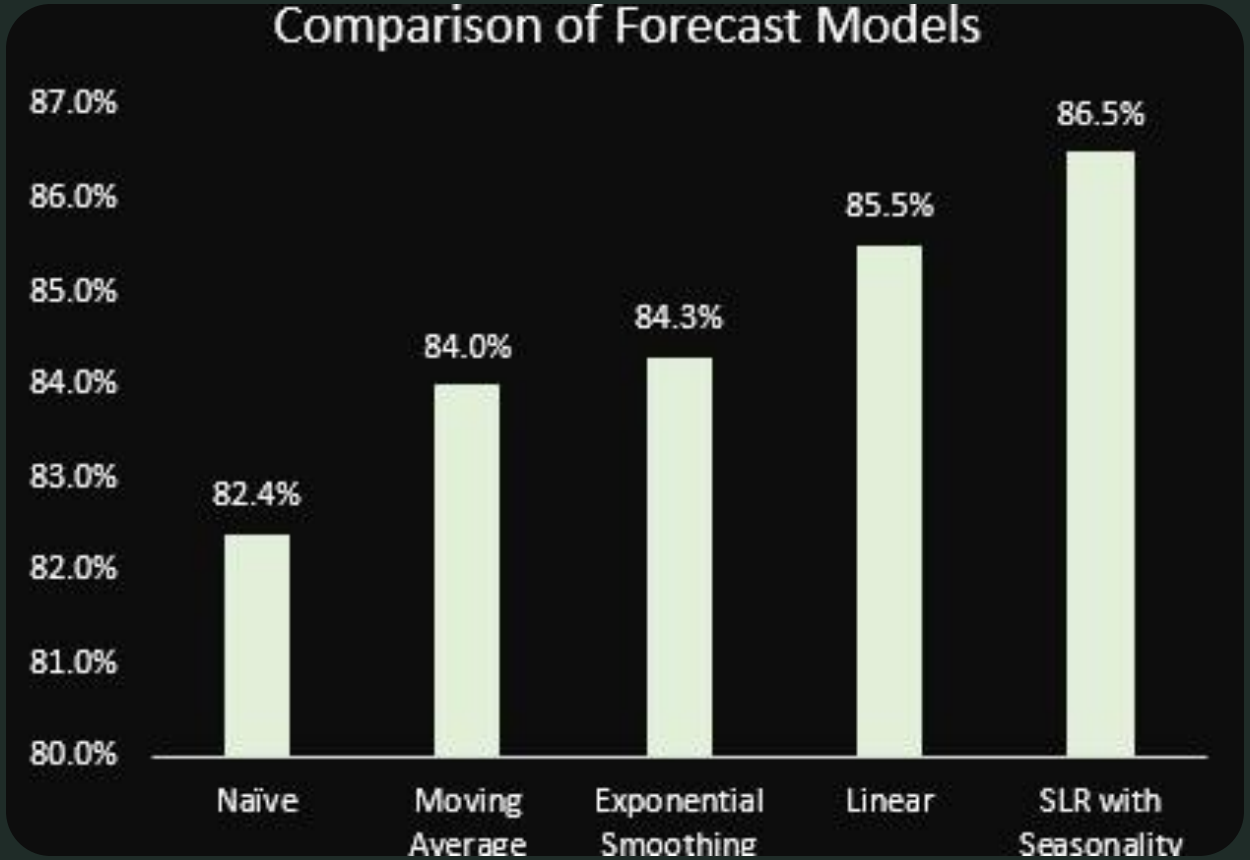
Goal

Explore various forecasting techniques and provide the management with the best recommendation.

Which models did we Test?

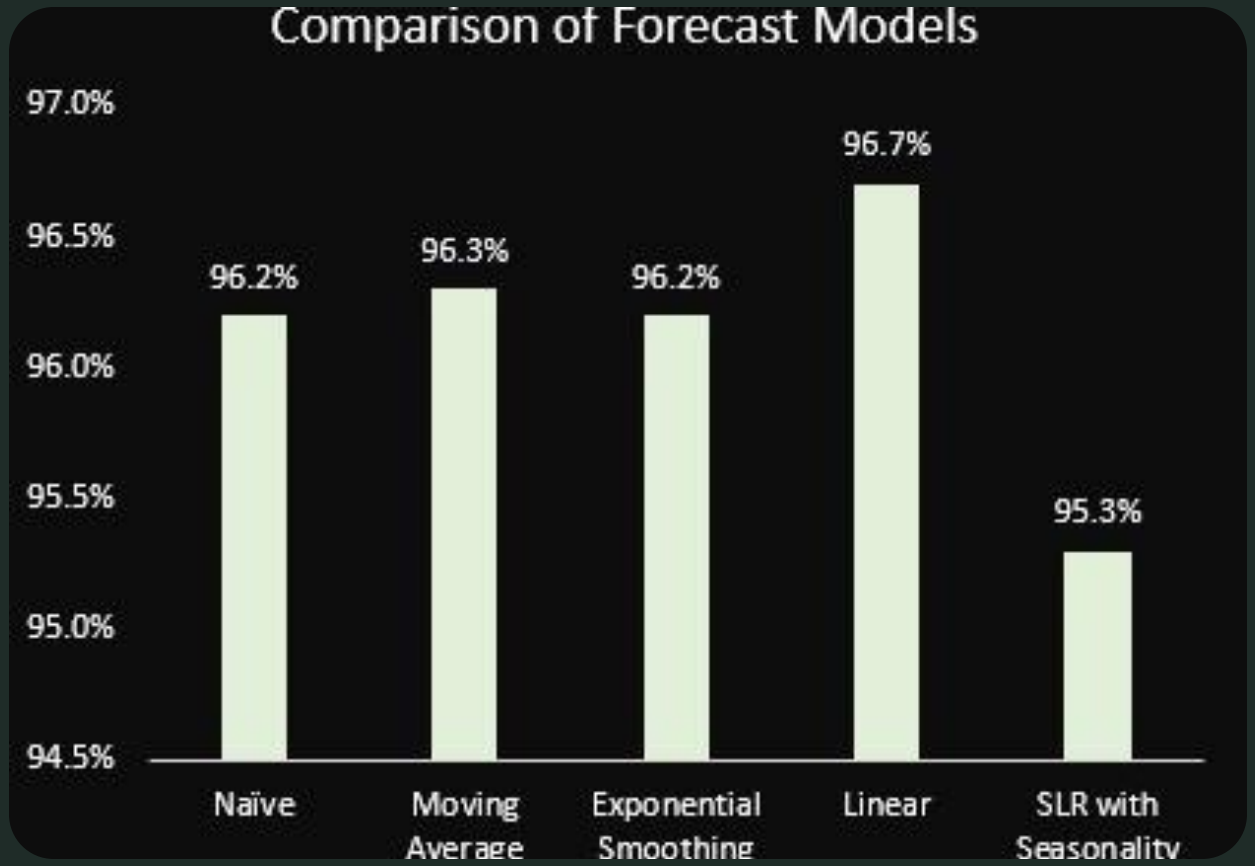
- **Naïve Forecast:** Assumes the next period is equal to the last.
- **Moving Average(2days and 2 months):** smooths recent fluctuations.
- **Exponential Smoothing**
- **Linear Forecast (Trend):** identifies long term direction.
- **SLR with Seasonality:** Accounts for repeating patterns.

Key Findings



Daily Sales: ★ SLR with Seasonality

- Naïve: 82.5%
- Moving average: 84%
- Exponential: 84.3%
- Linear: 85.5%
- **SLR with seasonality: 86.2% (Best)**



Monthly Sales: ★Linear (Trend)

- Naïve: 96.2%
- Moving average: 96.3%
- Exponential: 96.2%
- **Linear: 96.7%** (**Best**)
- SLR with seasonality: 95.3%

Why the Difference?

Monthly:

- The patterns show growth or decline
- It removes the daily ups and downs
- **Best Model: Linear Forecast**

Daily:

- The patterns repeat weekly
- Shows every ups and downs
- **Best Model: SLR (Seasonality)**

Daily Forecasting Models Comparison

Models	MAE	MSE	MAPE	ACCURACY
Naïve	7.27	84.22	17.15%	82.5%
Moving Avg (2 days)	6.55	66.46	16%	84%
Exponential Smoothing	6.49	3.25	15.7%	84.3%
Linear Forecast (Trend)	6.31	65.91	14.5%	85.5%
SLR with Seasonality	**5.63**	**50.41**	**13.8%**	**86.2%**

Monthly Forecasting Models Comparison

Models	MAE	MSE	MAPE	ACCURACY
Naïve	49.18	3355.73	3.8%	96.2%
Moving Avg (2 months)	49.95	3951.23	3.7%	96.3%
Exponential Smoothing	49.66	3639.83	3.8%	96.2%
Linear Forecast (Trend)	**42.43**	**2637.60**	**3.3%**	**96.7%**
SLR with Seasonality	60.22	4786.22	4.7%	95.3%

Recommendation

Daily or weekly Planning

Model: Simple Linear Regression with Seasonality.

It predicts short-term ups and downs based on pattern repetition.

Monthly Planning

Model: Linear Forecast (Trend)

It shows the growth or decline over time.

Thank you

Martha Maxwell
marthamaxwell02@gmail.com